How we went about this

A methodological account of the modular-degree project,
4 to 15 October 2026

Prepared in session `ccr-f42a94ab-alw7cu`
of the repository `dsr001001/mcptrial`,
at the request of the author, Dharmaj Soni

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Abstract

This document records how one person and one AI system went from a general brief, "search Cremona's tables of elliptic curves for relations that are not already known", to a 32-page research note with a theorem, a conjecture, and verification on three million curves, in eleven days. It is written for comparison with the accounts of other sessions that worked on the same material. It describes the plan, the decisions, the detours, the errors and how they were caught, and the discipline that held the work together. It does not restate the mathematics beyond what is needed to follow the story; the note itself and the repository do that.

1 The brief and the setting

The brief was item 9 of a longer list: symbolic regression across invariant databases of elliptic curves over $\mathbb{Q}$. The data source was John Cremona's `ecdata` repository, the upstream of the LMFDB tables, frozen at one git commit (`25cec5ec`) covering every curve of conductor below 500,000: 3,064,705 curves in 2,164,260 isogeny classes. The LMFDB web interface was not used; a pinned commit is a cleaner freeze and reproducible by anyone.

The division of labour was fixed from the start and never changed. The author set the goals, approved each plan before it ran, chose between options when they were presented, supplied outside opinions (referee reports produced by other systems or sessions), and took responsibility for the result. The AI system did the computing, the reading, the writing and the checking, and reported back after each step. Everything lives in one directory, `ecmine/`, on one branch, with a chronological log (`README.md`) and a resume document (`HANDOFF.md`) that any later session can read first.

2 The plan: ten steps, agreed before anything ran

The first decision was not to start computing. A ten-step plan was written and agreed:

1. Frozen snapshot of the data, with the exact commit recorded.
2. A flat feature table: everything the tables have, plus what PARI/GP can add (root number, reduction types and Kodaira symbols at bad primes, a_p for the 168 primes below 1000, j-invariant height, discriminant), with sanity checks against known identities (BSD to 10^{-5}, parity, Hasse bound, one optimal curve per class).
3. An exclusion list: every known identity, theorem, conjecture and heuristic relating pairs of columns, each exact relation verified on the table. The search space is the complement.

4. A null design, written as a pre-registration and signed off before any test ran: unit of analysis (isogeny class), fixed splits (working set $N \leq 300{,}000$; hold-out $(300{,}000, 400{,}000]$ untouched until step 8), nested-model tests (does X predict Y beyond what is already known about Y), a conditional permutation null, a hit criterion, a budget of 112 tests, and positive and negative controls.

5. Cheap models: run exactly the pre-registered tests.

6. Interpret the hits; identify known mechanisms.

7. Sharpen on the working set.

8. One test on the hold-out.

9. Literature check.

10. Write up.

The reasons were the usual ones for data mining in a domain where almost everything visible is already a theorem: without an exclusion list every hit is a rediscovery; without a hold-out every sharpened statement is overfitted; without controls there is no way to know whether the test can see anything at all. The last point mattered. One positive control (Watkins' bound on the modular degree) failed, because a variance-explained statistic cannot see an inequality constraint. Instead of lowering the bar, a second test family was added (forbidden, depleted and enriched cells of column pairs inside conductor strata) that can see constraints. The failed control is reported as failed in the step 5 report.

3 What the search found

Of 112 pre-registered tests, the negative controls were all clean and six of seven positive controls hit. Among the candidates, one was worth pursuing: U1, the ℓ-adic valuation of the modular degree of the optimal curve against local invariants at the bad primes (originally phrased against the Tamagawa product). Everything after this point is the story of one candidate.

Step 6 immediately identified the known mechanism: the multiplicative part at $\ell \geq 5$ is a theorem of Kim and Ota, building on Agashe, Ribet and Stein. The honest next question was whether anything in the data went beyond it. It did, in three ways that were stated as conjectures on 4 October and tested once on the hold-out: an additive contribution at $\ell = 3$ from Kodaira types IV and IV*, a "twisted" supplement from types II, II* and I_n^*, and a bound on the deficit in the Eisenstein case. Zero exceptions on 1.3 million curves of the working set and 430,000 of the hold-out.

4 The method of the middle phase: state, test, break, restate

From 4 to 10 October the work was a loop that ran perhaps a dozen times:

1. State the sharpest rule consistent with the working set.

2. Test it on data it has not seen: the hold-out, then the extension $(400{,}000, 500{,}000]$ with 359,009 further optimal classes, then 893 curves generated outside the tables with prescribed local types and conductor up to 3,994,600, whose modular degrees were computed from scratch with `ellmoddegree`.

3. When a rule survived, ask what mechanism would produce it, and look for a statement that the mechanism predicts and the current rule does not.

4. When a rule broke, find the cleaner statement that the break points to.

Two breaks were decisive. The first "all but the largest weight" rule for the twisted types was true on all the data and still wrong: separating the prime 2 from the odd primes showed it was an artifact of mixing them, and the clean rule (odd primes of type II, II* count 1 at $\ell = 3$, I_n^* counts $v_\ell(n)$, the prime 2 counts nothing through these types) replaced it. The second was the referee question "what are the exact local factors". Answering it meant computing, with PARI's `lfunsympow`, the Euler factor of the adjoint L-function at every bad prime, the wild primes 2 and 3 included, and discovering that the local term is the ℓ-adic valuation of the factor that the naive adjoint L-function of Diamond, Flach and Guo omits, plus a Tamagawa exponent at potentially multiplicative primes. Every earlier conjecture became a corollary of this one rule, and the equality rate at $\ell = 3$ rose from 39% to 57%, which is how one knows the rule is closer to the truth and not merely weaker.

A principle followed throughout: an inequality with zero violations is cheap; the equality rate, and where equality fails, carries the information. The calibration computed in the note (what fraction of bound-zero curves have $3 \mid \deg \varphi$, and the roughly 605,000 violations one would expect if the terms were independent) is what makes "zero violations" mean something.

5 Reports from outside, and what was done with them

On 10 October the author brought in four documents: a referee report written by an AI system, a research note assessing it, a second referee report, and an independent review that compared the submission to Diamond, Flach and Guo. The rule adopted, at the author's instruction, was: do not incorporate a suggestion before it has been assessed and discussed. In practice this meant:

- Every factual claim in a report was checked against the data before being believed. The claim that every exception of the uncorrected rule has mod-3 image the normaliser of a split Cartan (3Ns) was replicated exactly on $N < 100{,}000$ and then verified on all four ranges (475 exceptions, all 3Ns, shortfall always exactly 1). The reports' numbers matched ours, which is also a check on the reports.

- Every claim about the literature was checked against the source. Agashe, Ribet and Stein's Table 1 was read to confirm 99A1 has degree 4 and congruence number 12. The published text of Diamond, Flach and Guo (Annals 2004, pages 663 to 727) was extracted and read: Proposition 1.4(c), Theorem 3.7, Lemma 2.1, Theorem 2.7, the definition of η^Σ through the pairing. The arXiv long version 2512.02348 was identified as the same paper's expansion and not cited, because its numbering differs.

- Items were triaged into three groups (proof gaps, editorial, and the real mathematical question) and settled in that order, with the author choosing the order. The editorial items took one commit. The proof gap in the qualitative theorem (an appeal to "a congruence at almost all primes") was closed with two lemmas and a precise citation to Darmon, Diamond and Taylor. The mathematical question became a nine-page memorandum and then the restructured note.

The report that found the 3Ns pattern also offered an explanation (primes $q \equiv 1 \bmod 12$) that turned out to be wrong: the exceptions include $q \equiv 11 \bmod 12$ and $q = 2$. The correct description is image-based, not congruence-based. This is a case where checking the report's claim before adopting it changed the paper.

6 Reading the sources and assembling a proof

The evening of 10 October turned the rule from a conjecture into a theorem under the Taylor–Wiles hypothesis ($\ell \nmid 2N$, $\bar{\rho}$ absolutely irreducible on $G_{\mathbb{Q}(\sqrt{\ell^*})}$). The method was not to invent anything: it was to read Diamond, Flach and Guo with the specific question "what does their machine say about $v_\ell(\deg \varphi)$" and to write down the chain of their statements, with their numbering, that answers it. The chain has seven links, each a cited statement or a short lemma; the only new computations are the Tamagawa factor of the adjoint at a multiplicative prime (Lemma 2.7, with a sanity check against the Tamagawa number of the curve itself) and the identification of the correction term h_ℓ as the length of a local H^0 that vanishes exactly when Taylor–Wiles holds, which for $\ell = 3$ is exactly the 3Ns condition.

Each new lemma was then tested on the data before it went into the paper. The Tamagawa factor predicts an extra term exactly when $q \equiv 1 \bmod \ell$, $\ell \mid n$ and the unit part of the Tate parameter is an ℓ-th power. Computing Tate parameters for 46,222 curves at $\ell = 3$ gave zero violations and raised the equality rate among affected curves from 45% to 61%. A proof that also moves the equality rate in the predicted population is a proof one can trust.

7 Verification discipline

The habits that held throughout:

- **Independent recomputation.** The modular degree column was recomputed with PARI on fifteen curves spanning the configurations; the out-of-table degrees were all computed from scratch; a clean-clone reproduction (`repro_run.sh`) re-downloaded the data and rebuilt everything, and the summaries matched the committed ones exactly.
- **Nothing tuned on the hold-out.** The step 6 report says so in its second paragraph and it remained true.
- **Hand checks of proofs.** The determinant in the freeness argument of Lemma 2.9 of the revised note was computed by hand (-1). The exact degree formula of Remark 4.13 was checked numerically on three curves (11a1, 121a1, 539d1).
- **One-curve verification for the reader.** The note's appendix and the README give a PARI function `rule(E, l)` and Sage code so that any reader can check one curve in seconds. This was written early and kept current through every change of statement.
- **Counts, not adjectives.** Every "always" in the paper has a count beside it and a result file behind it.

8 Documentation and authorship

Four documents carried the project across sessions and across the author's own reading:

- `README.md`: a chronological log, appended to and never rewritten, with the commands that produced each result.
- `HANDOFF.md`: the state of the project, the mathematics as it stands with current numbering, the pipeline from a clean clone, the gotchas met, the conventions, and what could come next. Written so that a new session can resume "without much effort", which was the author's phrase.

- Dated snapshots: `Oct10/` (morning), `Oct10_v1_7pm/` (evening), `Oct15/` (the revised note from another session, with a review). Each has PDF, source where available, and a plain-text extraction so it can be read without a PDF viewer.
- The papers themselves, compiled PDFs committed so the author can read them anywhere.

Authorship and disclosure were settled on 10 October: the author's name, ORCID and email on every document; an unnumbered statement at the end of the note saying what the AI system did at every stage (data, code, proofs assembled from published statements, drafts, responses to reports), that the author could not have done it without the system, and that the author takes full responsibility for all statements, proofs, computations and errors. The author also ruled that no statement in the paper may defer to an outside review: the paper stands on its own and a journal decides.

9 Errors made, and how they were caught

A fair account lists them.

- The "all but the largest weight" rule (4 October) was an artifact; caught by separating $q = 2$ (6 October).
- The claim that type III exceptions have $q \equiv 1 \bmod 12$ (10 October morning) was wrong; caught by tabulating $q \bmod 12$ (140, 160, 72 for $q \equiv 1$, $q \equiv 11$, $q = 2$) the same evening.
- The finiteness of the correction term was first argued weakly through CM; replaced in the revision by a determinant argument proving $h_3 \leq 1$.
- The revised note says 28 of the 90 3Ns curves below 10,000 have $3 \mid N$; the count is 44. Caught by recomputation on 15 October; not yet fixed in the source.
- Bibliography entries added from memory (Zagier, Fontaine–Perrin-Riou, Hida, Flach, Tilouine) are flagged as such in `HANDOFF.md` for checking; the ones that could be checked (Diamond–Flach–Guo, Agashe–Ribet–Stein, Kim–Ota, Pasten) were.
- A remark in chat that the revised PDF contained em dashes was wrong (they are en dashes in page ranges). Recorded in `Oct15/OBSERVATIONS.md`.
- Tooling errors (a PARI member accessed as a Python attribute, `polgalois` on a reducible quartic, a valuation helper that ignored negative a_p) were caught by the checks they broke and are listed in the gotchas of `HANDOFF.md`.

The pattern is that every error was caught by a count, a hand check or a second look at the same data from a different angle, never by a report alone.

10 What worked, what did not

Worked.

- Pre-registration with controls. It cost a day and made the one candidate credible.
- One candidate, pursued to the end, rather than a list of twenty.
- Treating outside reports as hypotheses to test rather than instructions to follow.
- Reading the primary source with a specific question, and citing statement numbers.

- Testing each proved lemma on the data before publishing it.
- Snapshots and a handoff document, which made the 15 October review of a version prepared elsewhere a matter of an afternoon.

Did not, or cost time.

- The first positive control failed and a second test family had to be designed mid-step.
- Twice a rule was published in a snapshot and corrected the same day. Morning and evening versions on 10 October are the record.
- Bibliography entries from memory. A reference manager with the sources at hand would have avoided the flags.
- The revised note was prepared in another session without the repository's LaTeX source, so its three fixes cannot be applied here yet. Versions should flow through one source.

11 State on 15 October and open items

The live source is `paper/note.tex` (24 pages, 10 October evening). The revised version (32 pages, 15 October, `Oct15/note_v2_morning.pdf`) strengthens it: equality through a projection formula without multiplicity one, the Taylor–Wiles hypothesis automatic for $\ell \geq 5$, explicit Tamagawa factors at every reduction type, the 3Ns structure and the absence of 9-torsion proved. Three items stand before it becomes the live version: the count 28 should read 44; the identification in Section 4.1 of Diamond, Flach and Guo's integral structure with $H^1(X_0(N), \mathbb{Z}_\ell)_{\mathfrak{m}}$ needs a citation, a lemma or an explicit hypothesis; the AI-use statement needs rewording. After that: arXiv, then a journal (the recommendation on record is Journal de Théorie des Nombres de Bordeaux first, Journal of Number Theory second), and a Zenodo archive of the result files.

A Timeline by commit

Date	Commit	What
2026-10-04	2d671af	Frozen snapshot, feature pipeline (steps 1, 2)
2026-10-04	e931871	Exclusion list (step 3), null design draft (step 4)
2026-10-04	92a2bde	Step 5 harness, second test family after the failed control
2026-10-04	14b7040	Step 5 complete; report
2026-10-04	d5c399e	Steps 6 to 9 on U1: three conjectures, hold-out verified
2026-10-06	ec68583	Extension to 500,000, out-of-table generator, note draft
2026-10-06	4b64dc7	Twist-symmetrised bound replaces "all but the largest"
2026-10-10	3f26e31	Referee items 2 and 3: proof completed, editorial fixes
2026-10-10	1962d7a	Referee item 1: local-factor memorandum, Euler-factor rule
2026-10-10	00169a6	Note restructured around the rule; `Oct10/`; handoff
2026-10-10	7e502f9, 14389a0	Author details; AI-use statement
2026-10-10	1f4e045	Rule proved under Taylor–Wiles; `Oct10_v1_7pm/`
2026-10-15	f57c783	Review of the revised note; `Oct15/`

B Tools

PARI/GP 2.15.4 (`ellmoddegree`, `ellglobalred`, `lfunsympow`, `lfuneuler`, `ellinit` over $\mathbb{Q}_q$ for Tate parameters, `elldivpol`); Python 3 with pandas, numpy, pyarrow and cypari2; pdflatex with amsart and bibtex; pdftotext for the plain-text versions. All computations ran in the cloud

container of the session; the longest single run (Euler factors for the working set) took under an hour per shard in three parallel shards. Total compute for the whole project from a clean clone is a few hours.